

IBM University Engagement Project Submission

“Exploring the Power of Machine Learning with IBM Cloud”

Project Title – MachineGuard AI: Predictive Maintenance System for Industrial Machinery Using Machine Learning and IBM Watsonx AI

Domain of Project – Manufacturing Industry – Predictive Maintenance using Machine Learning

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Problem Statement -

Problem Statement No. 39 – Predictive Maintenance of Industrial Machinery

The Challenge

Develop a predictive maintenance model for industrial machines that can predict failures before they occur using real-time sensor data. The objective is to identify the type of machine failure (such as Tool Wear Failure, Heat Dissipation Failure, Power Failure, Overstrain Failure, or Random Failure) and recommend appropriate maintenance actions. This helps industries reduce downtime, improve machine reliability, and lower maintenance costs.

Proposed Solution -

MachineGuard AI is an intelligent predictive maintenance system developed using **Machine Learning, IBM Watsonx AI, and IBM Cloud Lite Services**.

The solution:

- Collects machine sensor inputs from the user.
- Uses a **Random Forest Classifier** trained on the Predictive Maintenance dataset.
- Attempts prediction through **IBM Watsonx AI**.
- Automatically switches to the **local Random Forest model** if the cloud service is unavailable.
- Displays:
 - Failure Type
 - Confidence Score
 - Severity Level
- Maintenance Recommendation
- Provides a user-friendly **Streamlit dashboard** for real-time prediction

Technology Used

IBM Cloud Services

- IBM Cloud Lite
- IBM Watsonx AI
- IBM Machine Learning
- IBM Watsonx Stdio
- IBM BOB

Programming Language-Python

Libraries

- Streamlit
- Pandas
- NumPy
- Scikit-learn
- Joblib
- Requests
- Python-dotenv

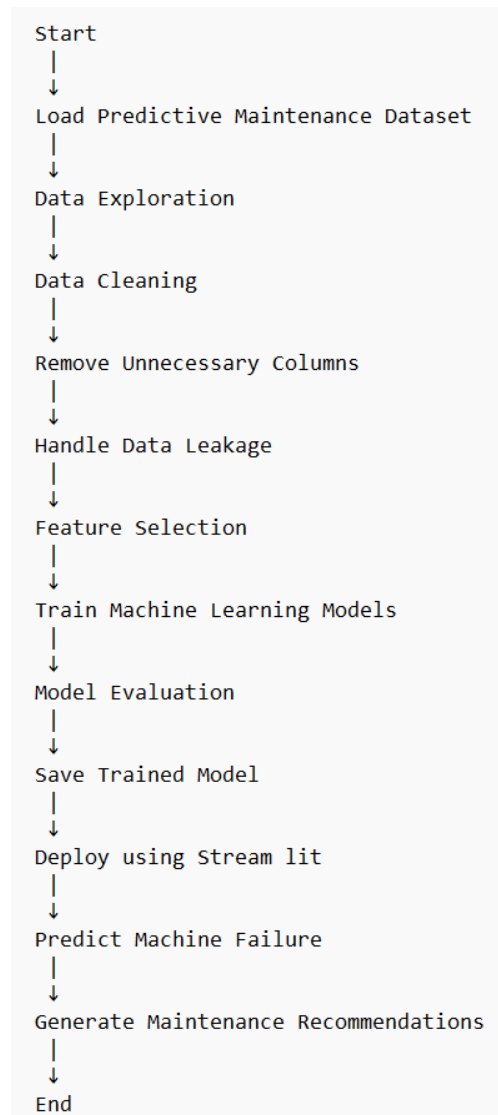
Machine Learning Model-Random Forest Classifier

Accuracy Obtained-97% (Training Accuracy)

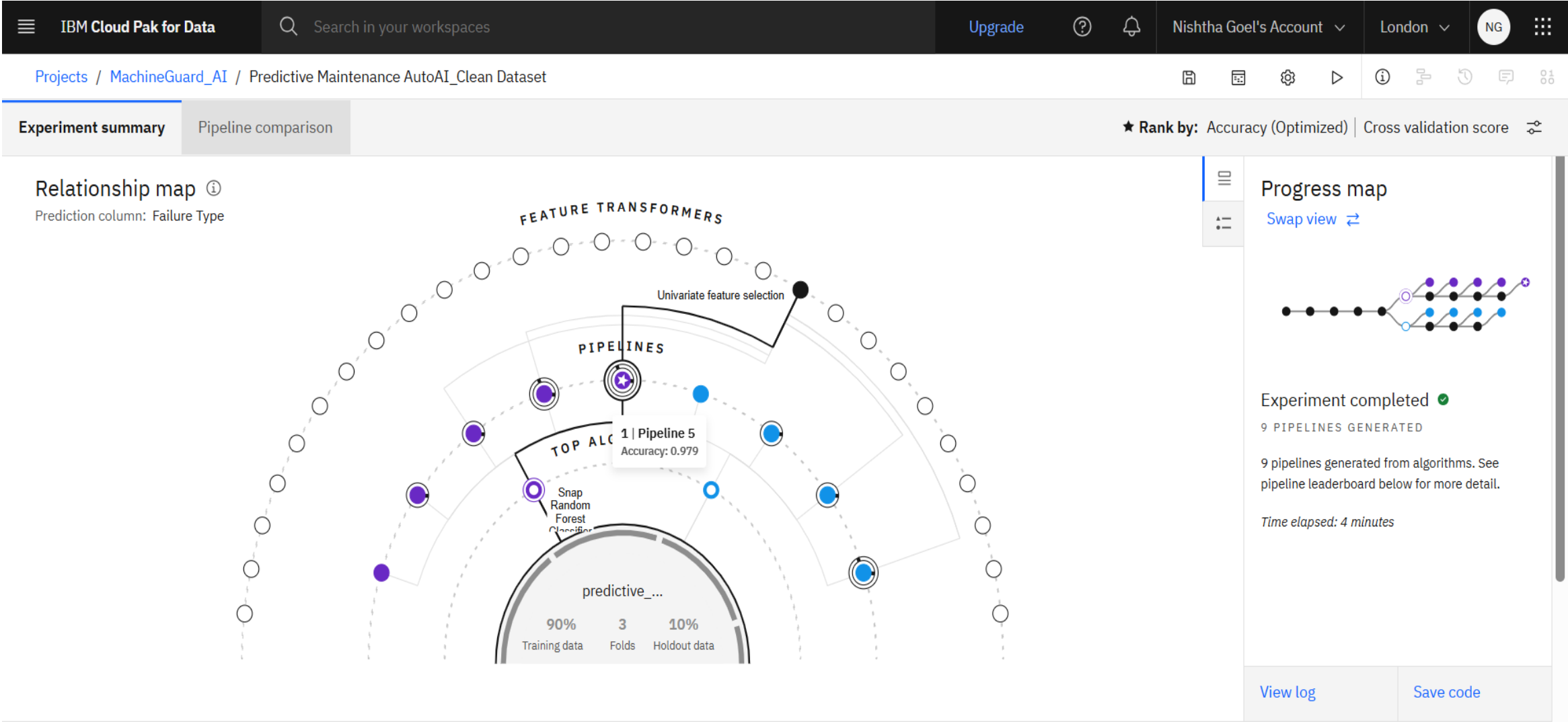
Enhancements

- Feature Selection
- Data Leakage Removal
- Local Model Fallback
- IBM Watsonx AI Integration
- Intelligent Maintenance Recommendation Engine

Workflow of the ML Project



Training Screenshots-



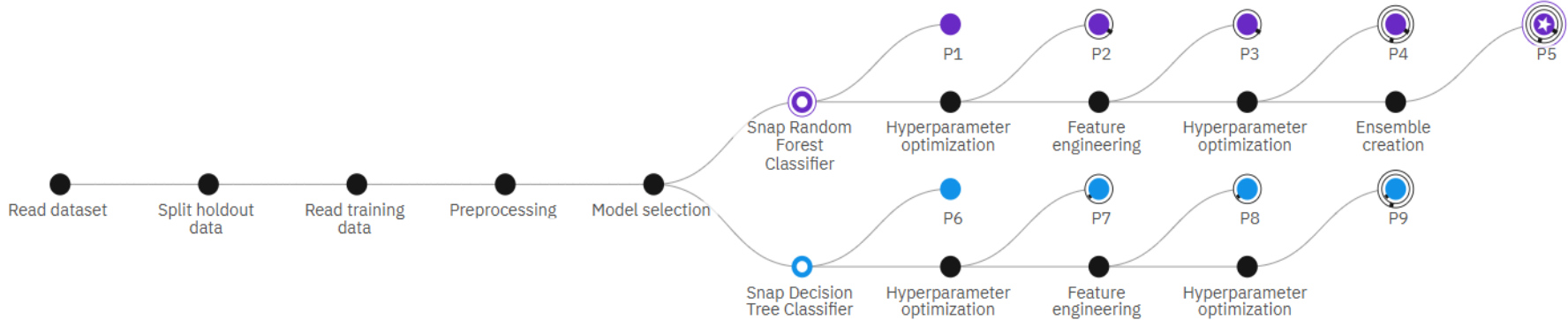
Experiment summary

Pipeline comparison

★ Rank by: Accuracy (Optimized) | Cross validation score 🔗

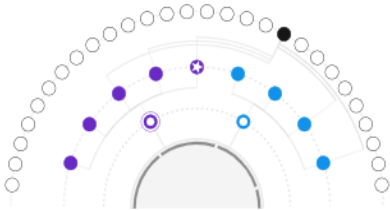
Progress map ⓘ

Prediction column: Failure Type



Relationship map

[Swap view](#) ↔️



Experiment completed ✓

9 PIPELINES GENERATED

9 pipelines generated from algorithms. See pipeline leaderboard below for more detail.

Time elapsed: 4 minutes

[View log](#)

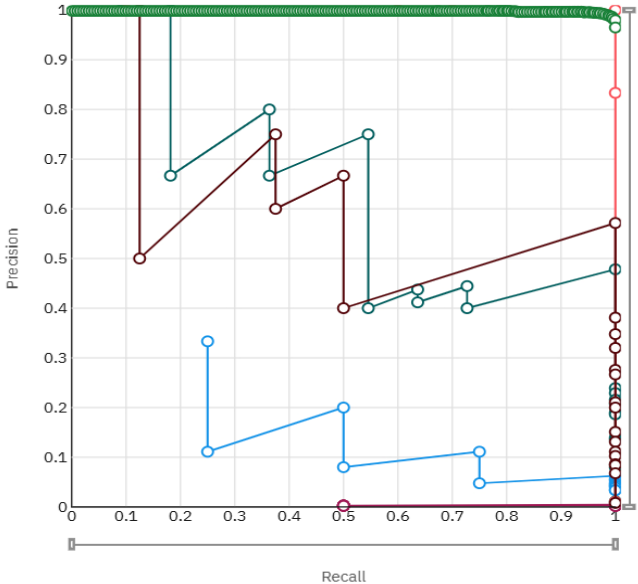
[Save code](#)

Best Model Screenshot:

Pipeline leaderboard 

	Rank 	Name	Algorithm	Specialization	Accuracy (Optimized) Cross validation	Enhancements	Build time	
★	1	Pipeline 5	 Batched Tree Ensemble Classifier (Snap Random Forest Classifier)	INCR	0.979	HPO-1 FE HPO-2 BATCH	00:00:53	
	2	Pipeline 4	 Snap Random Forest Classifier		0.979	HPO-1 FE HPO-2	00:00:49	
	3	Pipeline 2	 Snap Random Forest Classifier		0.978	HPO-1	00:00:06	
	4	Pipeline 1	 Snap Random Forest Classifier		0.978	None	00:00:02	
	5	Pipeline 3	 Snap Random Forest Classifier		0.978	HPO-1 FE	00:00:33	
	6	Pipeline 9	 Snap Decision Tree Classifier		0.977	HPO-1 FE HPO-2	00:00:04	
	7	Pipeline 7	 Snap Decision Tree Classifier		0.969	HPO-1	00:00:59	
	8	Pipeline 8	 Snap Decision Tree Classifier		0.968	HPO-1 FE	00:01:22	Save as
	9	Pipeline 6	 Snap Decision Tree Classifier		0.961	None	00:00:56	

Curves and Charts



Pipeline details
Pipeline 5

Rank
1

Accuracy (Optimized)
0.979 (Holdout)

Algorithm
Batched Tree Ensemble Classifier (Sn...

Specialization
INCR

Enhancements
HPO-1 FE +2

Save as

Model viewer

Model information

Feature summary

Evaluation

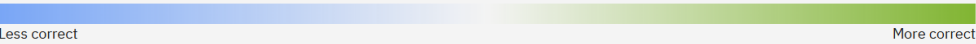
Model evaluation

Confusion matrix

Precision recall

Threshold

Observed	Predicted						Percent correct
	Heat Dissipation Failure	No Failure	Overstrain Failure	Power Failure	Random Failures	Tool Wear Failure	
Heat Dissipation Failure	3	8	0	0	0	0	27.3%
No Failure	0	964	1	0	0	0	99.9%
Overstrain Failure	0	6	2	0	0	0	25.0%
Power Failure	0	0	0	10	0	0	100.0%
Random Failures	0	2	0	0	0	0	0.0%
Tool Wear Failure	0	4	0	0	0	0	0.0%
Percent correct	100.0%	98.0%	66.7%	100.0%	0.0%	0.0%	97.9%



Pipeline details
Pipeline 5

Rank
1

Accuracy (Optimized)
0.979 (Holdout)

Algorithm
Batched Tree Ensemble Classifier (Sn...

Specialization
INCR

Enhancements
HPO-1 FE +2

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Model evaluation

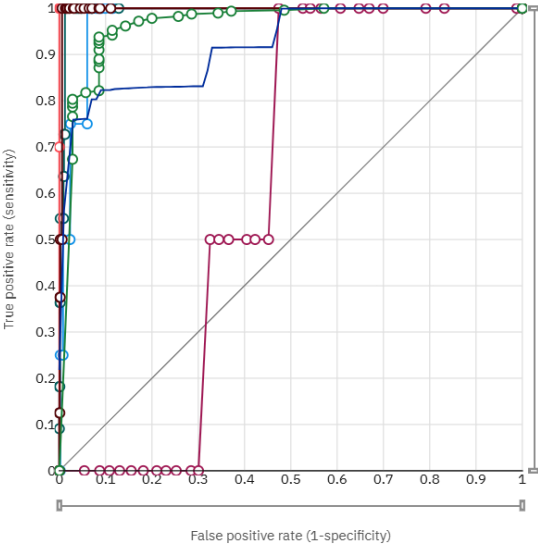
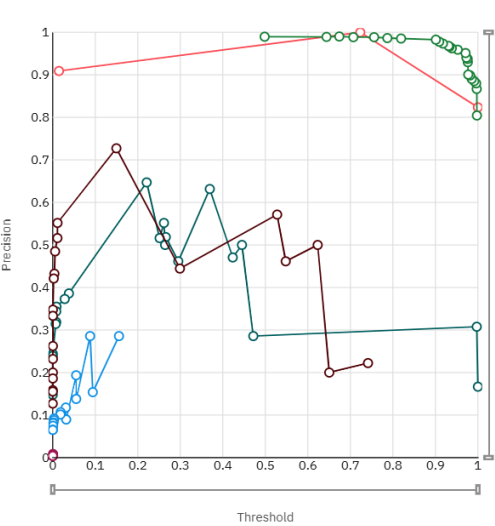
Confusion matrix

Precision recall

Threshold

Feature summary

All features			Search feature or transformer names
Feature name	Transformation	Feature importance	
NewFeature_2	pca(ALL)[2]	28.51%	
New_Feature_1_NSFA	None	25.17%	
Torque [Nm]	None	18.01%	
NewFeature_4	pca(ALL)[4]	15.07%	
NewFeature_0	pca(ALL)[0]	7.26%	
New_Feature_0_NSFA	None	5.98%	



Testing Screenshot:

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

?

🔔

Nishtha Goel's Account

London

NG

Deployment spaces / MachineGuard Deployment / P5 - Snap Random Forest Classifier: Predictive ... /

Predictive Maintenance Clean Model Deployment

✔ Deployed

Online

API reference

Test

Evaluations

Transactions

Enter input data

Text

JSON

Enter data manually or use a CSV file to populate the spreadsheet. Max file size is 50 MB.

Download CSV template

Browse local files

Search in space

Clear all

	Product ID (other)	Air temperature [K] (double)	Process temperature [K] (double)	Rotational speed [rpm] (integer)	Torque [Nm] (double)	Tool wear [min] (integer)
1	M14860	298.1	308.6	1551	42.8	0
2	L47230	298.9	309.1	2861	4.6	143
3	L47257	298.8	308.9	1455	41.3	208
4						
5						
6						

3 rows, 6 columns

Predict

Testing Screenshot:

Prediction results

Close X

Prediction type

Multiclass classification

Prediction percentage

3 records

No Failure

Power Failure

Display format for prediction results

☒ Table view

☐ JSON view

Show input data

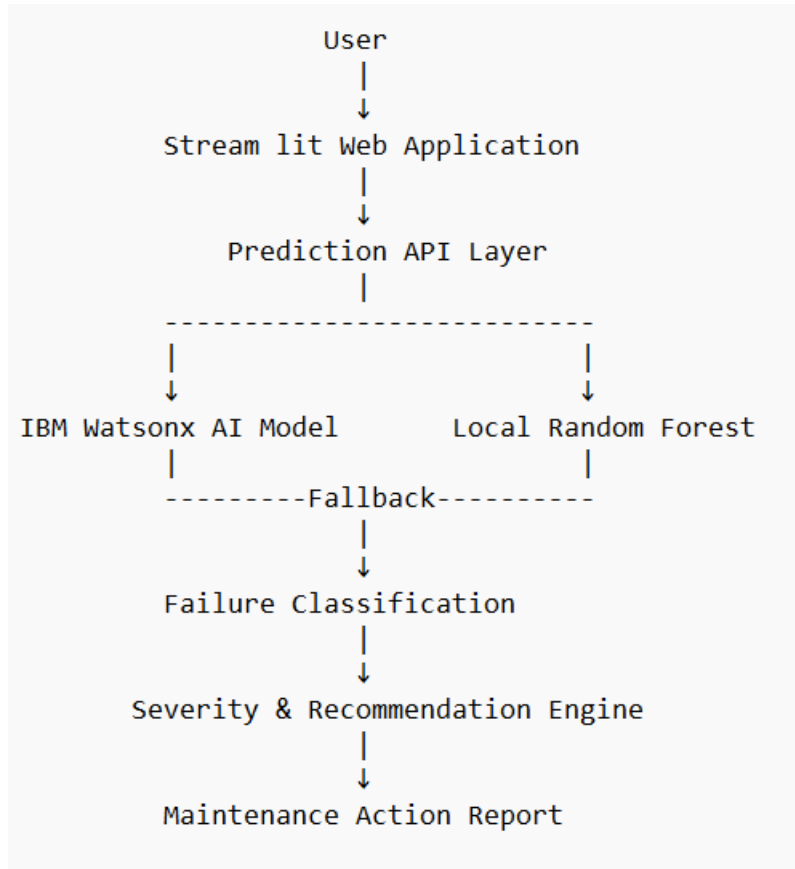
	Prediction	Confidence
1	No Failure	100%
2	Power Failure	100%
3	No Failure	93%
4		
5		
6		
7		
8		
9		
10		
11		

Download JSON file

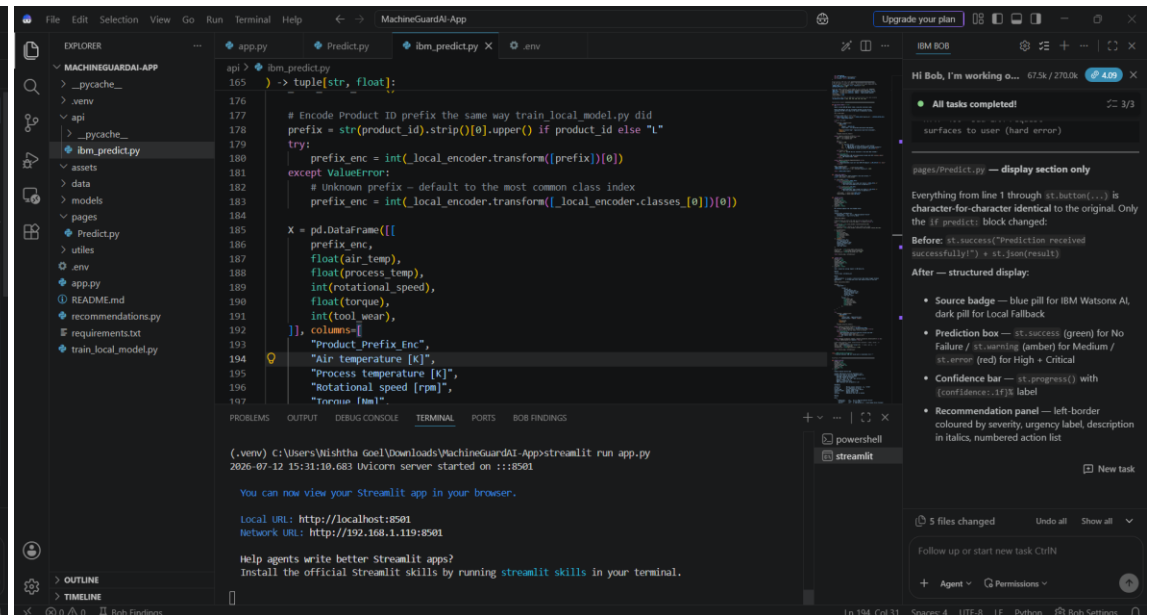
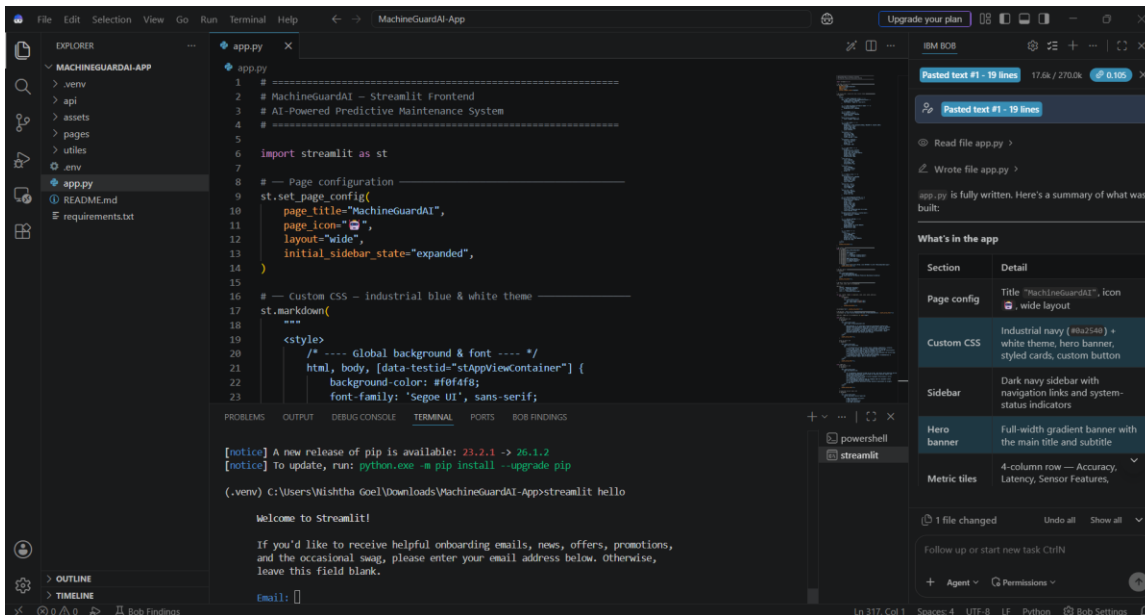
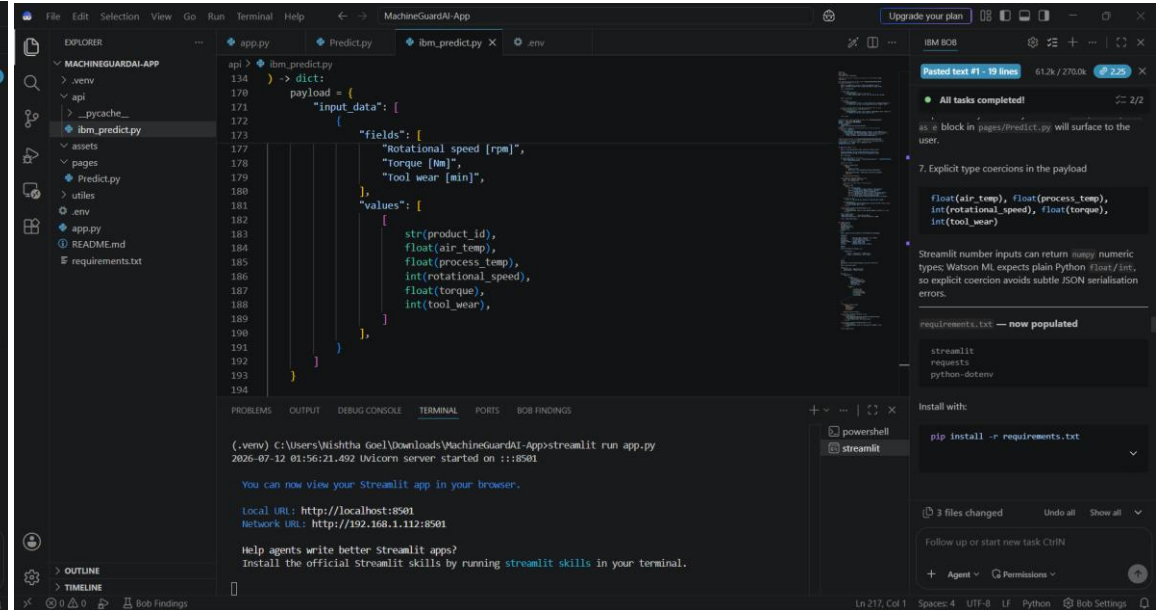
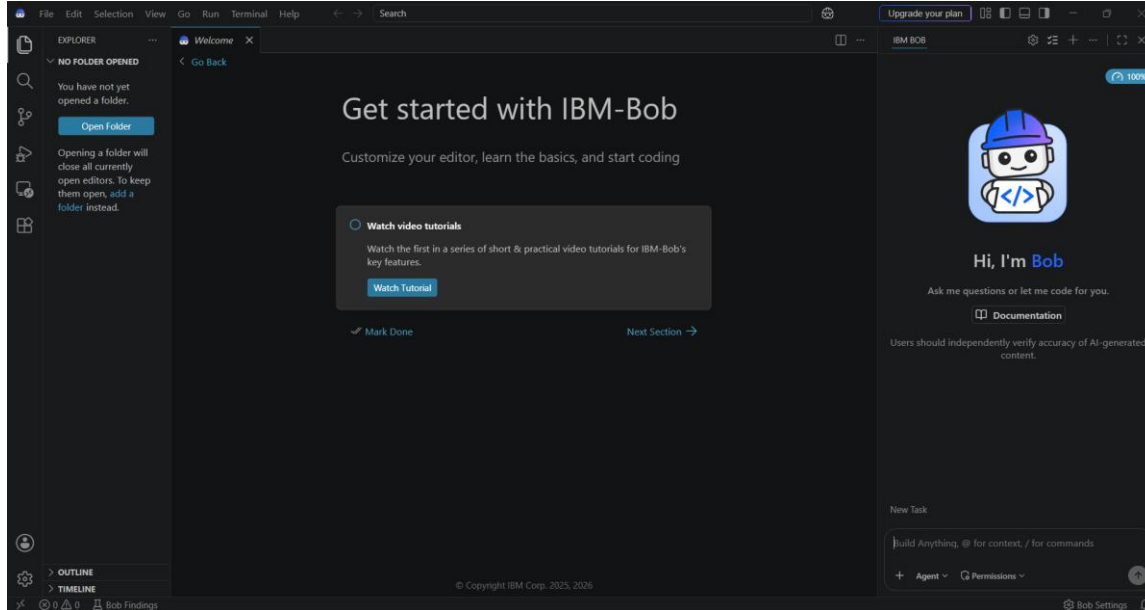
Required Files created using IBM BOB

- app.py
- recommendations.py
- train_local_model.py
- predict.py
- ibm_predict.py
- .env
- .venv

Architecture



IBM BOB Screenshots



Pages

- 🏠 Hello
- 📄 DataFrame demo
- ✓ Plotting demo
- 🌐 Mapping demo
- 🎞 Animation demo

Welcome to Streamlit! 🙌

Streamlit is an open-source app framework built specifically for machine learning and data science projects.

👉 Select a demo from the sidebar to see some examples of what Streamlit can do!

Want to learn more?

- Check out streamlit.io
- Jump into our [documentation](#)
- Ask a question in our [community forums](#)

See more complex demos

- Use a neural net to [analyze the Udacity Self-driving Car Image Dataset](#)
- Explore a [New York City rideshare dataset](#)

>>

Deploy

⌕ AI-POWERED PREDICTIVE MAINTENANCE

Prevent Downtime Before It Happens

MachineGuardAI analyses real-time industrial sensor data to detect early signs of equipment failure — giving your team the foresight to act before a breakdown occurs, reduce unplanned downtime, and maximise asset lifespan.

 Machine Learning

 Sensor Analytics

 Industrial IoT

Platform Capabilities









Project Output Screenshots

← → ↺ localhost:8501/Predict#predict-machine-health ☆ 🌙 🗂 📄 ⓘ

 **Predict Machine Health**

Enter the machine parameters below to predict equipment failures using MachineGuardAI.

Product ID

L47181

Air Temperature [K]

298.00

-

+

Process Temperature [K]

308.00

-

+

Rotational Speed [rpm]

1500

-

+

Torque [Nm]

40.00

-

+

Tool Wear [min]

100

-

+

Predict Failure

✓ UI Completed! IBM Prediction API will be connected next.

 **Overstrain Failure**

Confidence: 42.2%

Maintenance Recommendation

Severity: Critical | Urgency: Stop Operation

The product of tool wear and torque exceeds the mechanical strain threshold for the current machine variant. Continued operation will cause mechanical fracture, spindle seizure, or structural damage to the machine frame.

Recommended Actions:

1. STOP the machine immediately — do not continue the current operation.

2. Reduce feed rate and cutting depth before resuming any test runs.

3. Replace the current tool insert or cutting head; inspect for micro-fractures.

4. Review and reduce scheduled torque/load for this machine variant (L/M/H rating).

5. Inspect spindle bearings, chuck, and drive shaft for deformation.

✓ **No Failure** — Machine is operating normally.

Confidence: 99.0%

Maintenance Recommendation

Severity: None | Urgency: Routine

The machine is operating within normal parameters. No immediate intervention is required.

Recommended Actions:

1. Continue scheduled preventive maintenance intervals.

2. Log current sensor readings as a healthy baseline.

3. Verify lubrication levels are within specification.

4. Confirm next planned maintenance date is on record.

← → ↺ localhost:8501/Predict ☆ 🌙 🗂 📄 ⓘ

Maintenance Recommendation

Severity: Critical | Urgency: Stop Operation

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3. Replace the current tool insert or cutting head; inspect for micro-fractures.

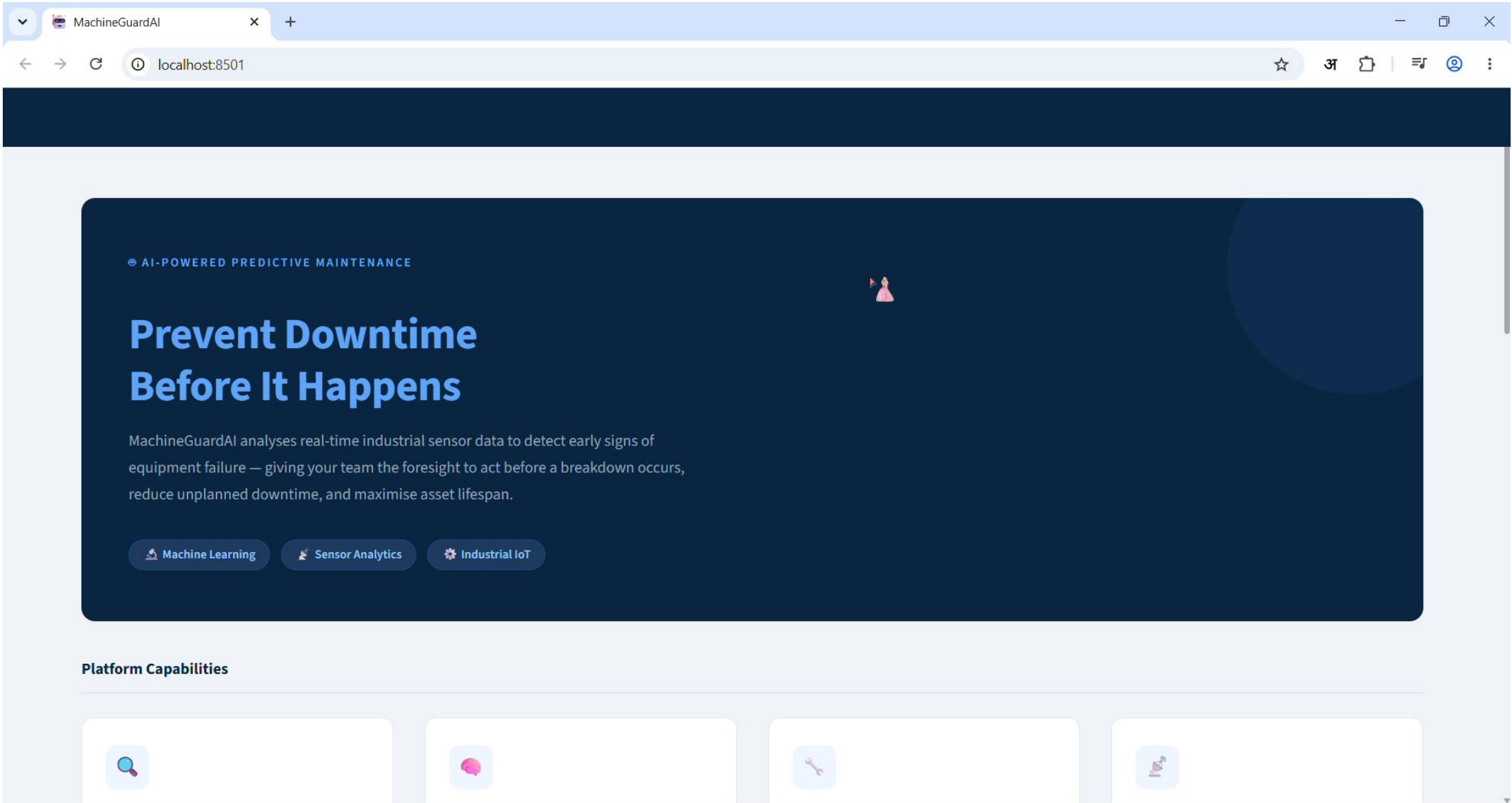
4. Review and reduce scheduled torque/load for this machine variant (L/M/H rating).

5. Inspect spindle bearings, chuck, and drive shaft for deformation.

6. Perform a full mechanical inspection of the gearbox and coupling.

7. Update machining program with corrected load parameters before restart.

Project



Git Hub Link

Project Repository & Live Application

GitHub Repository

The complete project source code, trained machine learning models, documentation, project report, and application files are available on GitHub.

☞ Repository Link

<https://github.com/Nishtha-tech12/MachineGuardAI>

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Live Streamlit Application

The MachineGuard AI application has been successfully deployed on **Streamlit Community Cloud**, allowing users to test machine failure prediction in real time.

Live Demo

<https://machineguardai-e5qyic3cemuabhdrxu3oam.streamlit.app>

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Novelty and Uniqueness

- **Hybrid Prediction System** – Integrates **IBM Watsonx AI** with a **locally trained Random Forest model** for reliable predictions.
- **Automatic Fallback Mechanism** – If IBM Cloud services are unavailable, the application seamlessly switches to the local model, ensuring uninterrupted operation.
- **Intelligent Maintenance Recommendations** – Provides severity-based maintenance actions instead of only predicting failure types.
- **Interactive Streamlit Dashboard** – User-friendly interface for real-time machine health prediction and monitoring.
- **IBM Technologies Integration** – Developed using **IBM Cloud Lite**, **IBM Watsonx AI**, and supported by **IBM BOB** during project development.
- **Real-World Industrial Application** – Designed to support predictive maintenance, reduce downtime, and improve operational efficiency in manufacturing environments.

Future Scope

- Integrate **IoT sensors** for real-time machine data collection.
- Enable **live monitoring dashboards** for continuous equipment health tracking.
- Improve prediction accuracy using **Deep Learning** and **Time-Series Models**.
- Implement **automated maintenance alerts** via email, SMS, or mobile notifications.
- Develop **AI-based maintenance scheduling** to optimize service intervals.
- Integrate with **ERP, MES, and Industrial Control Systems** for enterprise-wide deployment.
- Build **mobile and cross-platform applications** for remote monitoring and maintenance management.
- Expand cloud deployment to support **large-scale industrial environments** with scalable AI services.

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Certificates Earned:

Credly certificate



Certificates Earned:

Credly certificate



Certificates Earned:

IBM BOB Certificate

IBM SkillsBuild	Completion Certificate
------------------------	------------------------



This certificate is presented to

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(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 29 Jun 2026 (GMT)	Learning hours: 30 mins
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Certificates Earned:

Exploring Quantum Computing Certificate

IBM **SkillsBuild**

Completion Certificate



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Exploring Quantum Computing

(ALM-COURSE_3825253)

According to the Adobe Learning Manager system of record

Completion date: 11 Jul 2026 (GMT)

Learning hours: 50 mins

Certificates Earned:

Level up Cybersecurity with Generative AI Certificate

IBM **SkillsBuild**

Completion Certificate



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Level Up Cybersecurity with Generative AI

(ALM-COURSE_3947352)

According to the Adobe Learning Manager system of record

Completion date: 29 Jun 2026 (GMT)

Learning hours: 1 hr 30 mins

Thank You!

Thank you for your time and interest.